

Opening remarks

What the application does

Course discovery

Evaluations and attribution

Profiles, memories, and history

Schedules, calendar, and audits

Chat and natural-language edits

How the application works

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Output validation (civility / policy)

Agent Overview

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Atlas

AI-Assisted Schedule Builder for JHU Undergraduates

Alina Pan Jennifer He James Guo Junzhe Shi
Rachael Pei

Johns Hopkins University
Final presentation

May 5, 2026

One-liner: Natural-language course discovery, personalized schedules, and agentic planning—not a static catalog search.

Opening: Problem, Users, Why AI

Problem

JHU students need to discover courses, respect constraints, and build feasible schedules using SIS data and evaluations—without tab-hopping and guesswork.

Users

Undergraduates planning a term: search, shortlist, compare workload, and iterate with advising-style help.

Why AI is essential (not bolted-on)

- ▶ **Iter 1:** Semantic search + match explanations over embeddings; agent orchestrates tools.
- ▶ **Iter 2–3:** Onboarding → structured memories; schedule-aware chat; NL add/drop/swap.
- ▶ **Iter 4:** Clarification-gated mutations; unified search; parallel audits + quality gates.

Speaker: @Alinapanyue

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Demo 1 — Course discovery

Goal

Open-ended query → ranked results; unified search with `matchType`; optional disambiguation (Iter 4 R3).

- ▶ Show: natural language query → results list + explanations where semantic.
- ▶ Mention: `searchCourseDescriptions` (Iter 1) → orchestration (Iter 2 R6) → clarification state (Iter 4).

Speaker: @rachael-p Time: ~4 min

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Demo 2 — Eval summaries + attribution

Goal

On-demand eval summary grounded in metrics; raw/source table (Iter 4 R1).

- ▶ Show: “Summarize evals” on a course card → narrative + metrics.
- ▶ Open **raw** / **source** modal: terms, instructors, numeric rows aligned to claims.
- ▶ If no data: UI must not fabricate (Iter 1 R4).

Speaker: @JunzheShi0702 Time: ~4 min

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Demo 3 — Onboarding, memories, history

Goal

Preferences from onboarding + chat → long-term memories; completed courses / transcript path (Iter 4 R2).

- ▶ Show: onboarding or profile → `derived_memories` / memory list.
- ▶ Optional: completed-course history or transcript review flow (matched / ambiguous / unmatched).

Speaker: @James-Guo-03 Time: ~4 min

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Demo 4 — Schedules, calendar, audit

Goal

Dashboard → schedule page → workload audit; list vs. weekly calendar (Iter 4 R4).

- ▶ Run audit; show feasibility / workload narrative.
- ▶ Toggle **list** / **calendar**; mention parallel checks + synthesis (Iter 4 R5) at high level.

Speaker: @chjenniferhede Time: ~4 min

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Demo 5 — Schedule-aware chat + NL edits

Goal

Chat with schedule context; NL add/drop/swap via `modifyScheduleCourses`; clarification if ambiguous (Iter 3 R1, Iter 4 R3).

- ▶ One concrete NL command that mutates the schedule on deployed app.
- ▶ If time: trigger or describe clarification loop (saved state per conversation).

Speaker: @rachael-p (or handoff with @chjenniferhede for chat UI) Time: ~4 min

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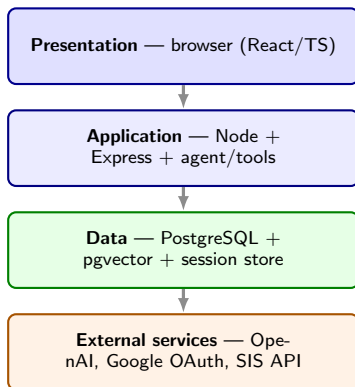
Questions

Stack: Logical Layers

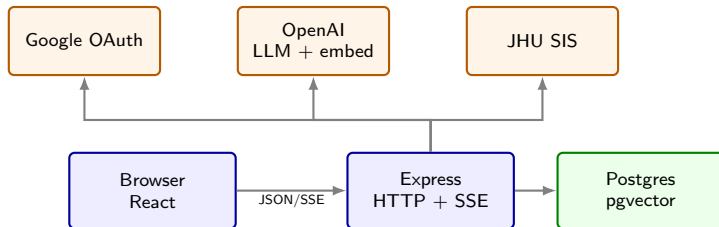
Responsibilities

- ▶ **Client:** React + Vite, streaming UI, schedule views
- ▶ **API:** Express routes, auth session, agent endpoint
- ▶ **Data:** Postgres + pgvector, chat/audit persistence
- ▶ **External:** OpenAI, Google OAuth, JHU SIS

Speaker: @JunzheShi0702



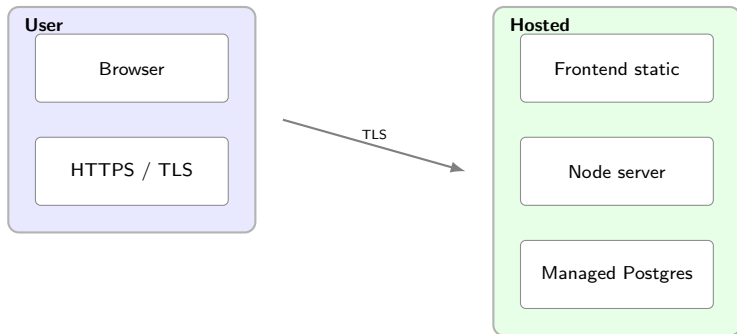
Stack: Components



Integration notes

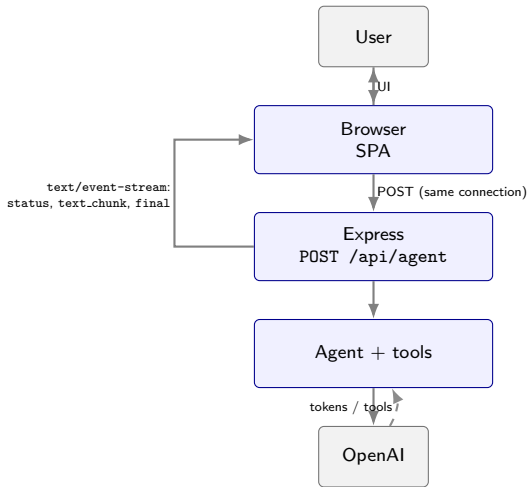
Session cookie authenticates dashboard routes; `/api/agent` upgrades to SSE for streaming tokens and tool progress (Iter 3 R6).

Deployment (Typical)



Swap placeholder host names on the right for your real stack (e.g. Vercel + Railway + Supabase).

Runtime: chat request and SSE

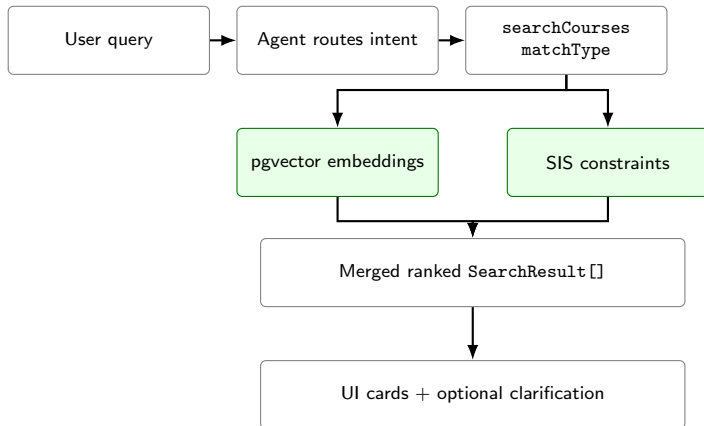


Persistence (Iter 3)

User message saved immediately; assistant text on final; partial + aborted metadata on stop.

Speakers: @rachael-p, @Alinapanyue

Data flow: unified course search



Iter 4 R3: one merged search surface; semantic vs. exact vs. hybrid encoded per row.

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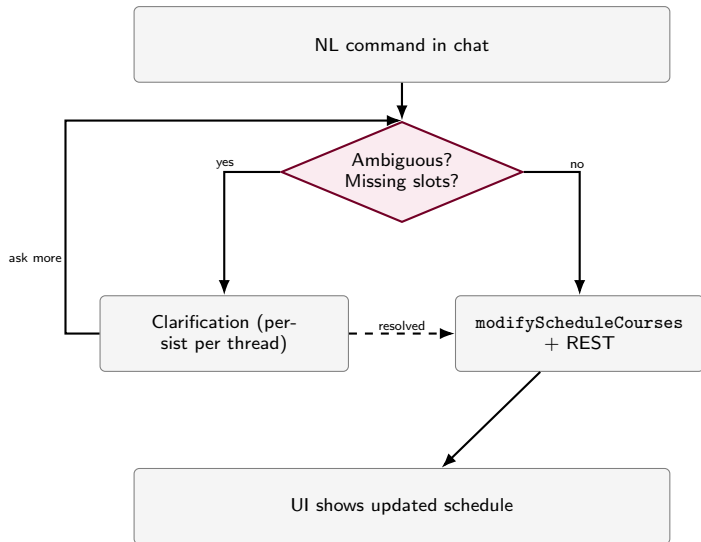
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Control flow: NL schedule edit



Iter 3 R1 + Iter 4 R3: block mutations until required clarification slots are confirmed.

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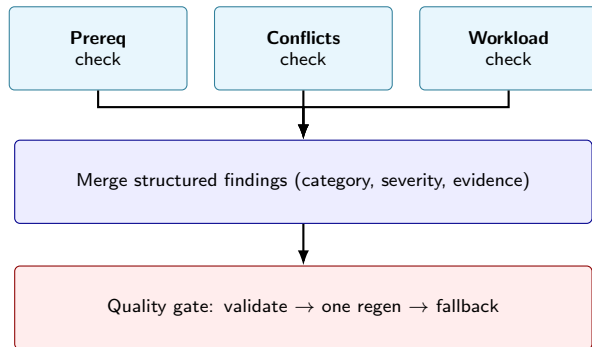
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Parallel Audit Synthesis and Gate



Failure isolation (Iter 4 R5)

One failed check still returns partial findings; gate telemetry (`gate_pass`, `gate_fail`, ...) where implemented.

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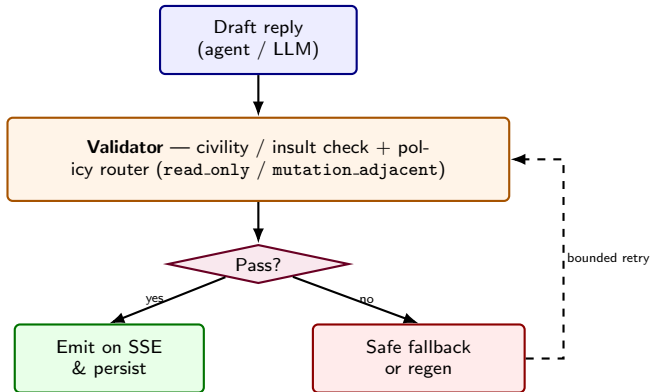
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User-visible text: validator before stream (Iter 4)

Assistant text can occasionally include **insulting** or **irrelevant** contents. A **validator** runs on draft output *before* chunks are committed to the SSE stream.



Same family as quality gates (Iter 4 R5)

Treat civility like other **hard failures**: validate → at most one constrained regen → **safe fallback** if still failing; log `gate_fail` / policy reason for observability.

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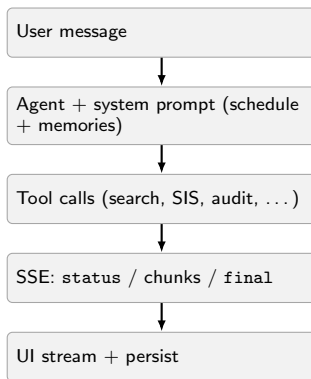
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Agent Loop

Tooling (Iter 1–4)

- ▶ Models: e.g. gpt-4o-mini; embeddings
- ▶ text-embedding-3-small
- ▶ Search, SIS details, eval summary, metrics, audits, schedule mutate
- ▶ Unified searchCourses; clarification state before writes



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Testing, AI eval, observability

Non-AI (backend/ + frontend/)

- ▶ `npm run test` in backend/ and frontend/ (Vitest).
- ▶ `frontend/: npm run test:e2e` (Playwright).
- ▶ CI and unit/E2E expectations per final delivery rubric.

AI evaluation

- ▶ Golden prompts / structured checks; agent regression paths.
- ▶ Gate telemetry where shipped (Iter 4 R5).

Coverage

- ▶ `npm run test:coverage` in backend/ and frontend/ for line/branch reports (Vitest + v8).
- ▶ Tie spoken numbers to what CI actually runs; note gaps honestly at Q&A.

Speakers: @James-Guo-03 + @chjenniferhede

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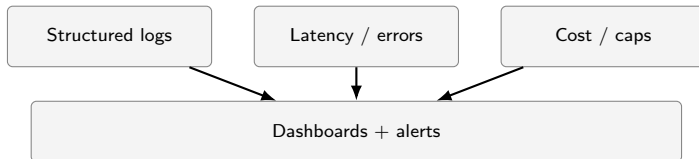
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Ops feedback loop (conceptual)



Final delivery rubric: tie spoken claims to what is actually deployed (logging, rate limits, validation). Mark items as **shipped** vs. **planned** live.

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Closing: Limitations, future work, reflection

Limitations (be honest)

- ▶ Partial eval coverage; SIS rate limits / stale sections; edge cases in clarification and gates.
- ▶ Model failures: fallbacks and user-visible errors still evolving.

Future work

Stronger observability dashboards; broader course history; tighter prompt/tool contracts; more automated AI evals.

Reflection

Team coordination across four iterations; using AI tools as dev partners; trade-offs (latency vs. depth, autonomy vs. safety).

Speaker: @James-Guo-03 (different from opening)

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Thank you — Questions

Questions?

Please ask questions about the project, the team, the process, the technology, the future, etc.